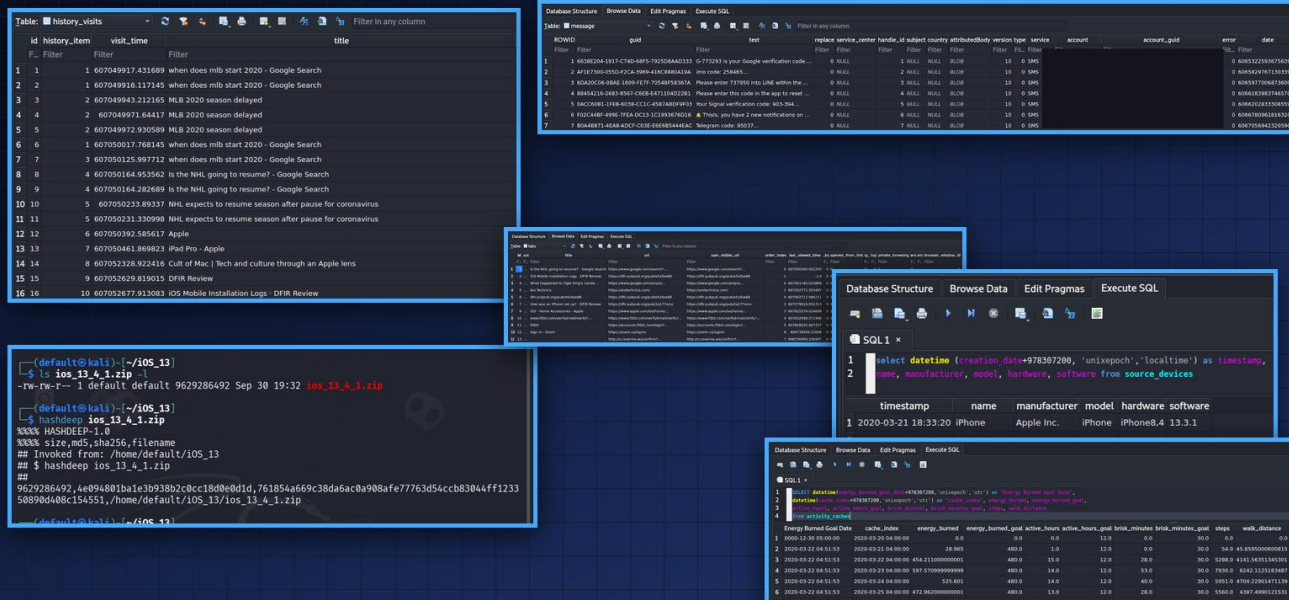




CIP-B106 | MOBILE AND IOT FORENSICS

Digital Forensic Examination Report

iPhone iOS 13.4.1 file-system extraction: device profile, artefact findings and event chronology

Case identifier	CIPB106-iOS13-C11/26/DFIT/17310
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Cohort	COHORT 11
Submission date	4 TH October 2026
Evidence examined	iOS 13.4.1 training image (ios_13_4_1.zip, 13-4-1.tar file-system extraction)

Academic Integrity Declaration

I confirm that this report reflects my own examination of the authorised course evidence and that I have identified all sources, tools, and assistance used.

Signed: EBENEZER ELIKEM AMANKWAH OFORI DZAM Date: 04/10/2026

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2 Executive Summary

This report examines the iOS 13.4.1 training image issued for Case CIPB106-iOS13 using a verified working copy, DB Browser for SQLite, plistutil and command-line tools. The package matched the expected MD5, and its SHA-256 matched the full 64-character value recorded locally (the brief prints 63 characters). The Health records show an iPhone8,4 running iOS 13.3.1 and later 13.4.1, paired with an Apple Watch, AirPods and AirPods Pro; Photos metadata indicates the America/New_York time zone. sms.db holds automated verification messages from services whose apps appear in the SpringBoard layout; the voicemail table is empty. A 36-hour chronology (27 to 28 March 2020 UTC) links Health, call, message, contact and Safari records, with two Safari relationships corroborated across three databases. Principal limitations: KnowledgeC, location rows, calendar content, SIM and account identifiers, and jailbreak indicators are not covered by the exhibits compiled here; Health times used the examiner machine's local time; call and message direction was not confirmed. Findings describe device artefacts, not who used the device or why.

3 Evidence and Method

3.1 Evidence register and handling

The examination used one primary item. The compressed training package was retrieved with wget into a dedicated working folder (~iOS_13) on a Kali Linux examination workstation on 30 September 2026, hashed, and only then unpacked. The embedded 13-4-1.tar file-system archive was extracted with tar -xvf to a separate directory tree. The downloaded ZIP was never opened for writing, renamed or re-uploaded. The iTunes backup and Sysdiagnose logs described in the brief are not referenced in the exhibits compiled for this draft.

Item	Detail	Status
ios_13_4_1.zip	Size 9,629,286,492 bytes Working folder ~iOS_13; hashed with hashdeep	Hash verified (Section 3.2)
13-4-1.tar (inside "iOS 13.4.1 Extraction/Extraction")	Extracted with tar -xvf to Volumes/JOSH/NoTar-13-4-1/	Extracted; tree listed in Figure 3
Examination workstation	Kali Linux VM; DB Browser for SQLite, plistutil, sqlite3, hashdeep, tar, file manager	2026
iTunes backup; Sysdiagnose logs	Part of the course package	Not referenced in these exhibits

Handling statement. Databases were opened in DB Browser for SQLite for browsing and SELECT queries, and no Write Changes action is recorded in the exhibits. One sqlite3 export (Appendix C) was run without the -readonly option; opening a write-ahead-logged database this way can create or checkpoint -wal and -shm files in the working copy, so the original evidence is unaffected but the working copy should be re-hashed or re-extracted before further use. Telephone numbers, e-mail-format accounts, verification codes and GPS coordinates are masked in the narrative and redacted in figures.

3.2 Integrity check

hashdeep reported MD5 4e094801ba1e3b938b2c0cc18d0e0d1d for the 9,629,286,492-byte package (Figure 2). This is identical to the MD5 given in the brief. The SHA-256 reported on the workstation is 761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551 (64 hexadecimal characters). The SHA-256 printed in the brief is one character shorter (63 characters) and omits a "4" near the end ("...0d408c15551" against "...0d408c154551"). All other characters agree. The shorter value cannot be a valid SHA-256 digest, so the discrepancy is most likely a typing error in the brief.

3.3 Tools and approach

Each task started from the file system: paths were located in the extracted tree and databases and property lists were read directly. DB Browser for SQLite handled browsing and SQL queries, plistutil converted binary plists to XML, and sqlite3 produced one export. No commercial forensic platform appears in the exhibits, so every finding here is a manual finding from a file, database or plist. No tool-versus-manual comparison is therefore available (Appendix E).

3.4 Timestamp conversion method

Most iOS databases store Apple Cocoa Core Data time: seconds since 2001-01-01 00:00:00 UTC. A Unix timestamp is obtained by adding 978,307,200 seconds. From iOS 11 onward, sms.db message.date holds nanoseconds, so the value is divided by 10^9 first. Example: sms.db message ROWID 1 raw date 606532259367563904 ns gives 606,532,259.37 s, plus 978,307,200 gives Unix 1,584,839,459, which is 2020-03-22 01:10:59 UTC.

All times in this report are UTC unless labelled, with Eastern Daylight Time (UTC minus 4 hours) shown beside key events. The device time zone is supported by the ZTIMEZONENAME value America/New_York in Photos.sqlite and by US east-coast call locations; daylight saving time began on 8 March 2020, so EDT applies to every date used. The activity_caches cache_index values fall at 04:00:00 UTC each day, which is local midnight at UTC minus 4 hours, so Health independently agrees with the Eastern Daylight Time offset. Whether the device clock was set automatically was not examined, so converted times are qualified accordingly.

4 Device and Account Profile

4.1 Operating system, model and paired devices

The health data source table (Figure 24) records the handset as hardware identifier iPhone8,4, listed by Apple as the first-generation iPhone SE. It appears twice, with software 13.3.1 (first row timestamp 2020-03-21 18:33:20, examiner-local) and 13.4.1 (2020-04-13 15:26:44, examiner-local). That is consistent with the iOS 13.4.1 build named in the brief and with an operating-system update between those dates. The same table lists an Apple Watch (identifier Watch4,3) at watchOS 6.1.3 and later 6.2.1, AirPods (hardware 0x2002) first recorded 2020-03-27 15:06:14 and AirPods Pro (0x200e) first recorded 2020-03-31 15:08:08 (all examiner-local). The meaning of the creation_date column is the time the Health database recorded the source, not the time the hardware was purchased or paired.

Serial number, build string, ICCID, IMSI, MCC and MNC: these require LastBuildInfo.plist, com.apple.commcenter.plist and general.log (Appendix D)

Identifier classes matter for interpretation. The telephone number (MSISDN) and IMSI belong to the subscriber and SIM; the ICCID identifies the physical SIM card; MCC and MNC identify the home network operator; the serial number and hardware identifier belong to the handset. Only the handset hardware identifier is evidenced so far.

4.2 Subscriber and account evidence

sms.db records the device's own SMS account as a United States number in the 919 area code (digits masked; Figure 13). The chat table also contains one iMessage chat (ROWID 13) whose account login is an e-mail-format identifier, separate from the SMS account (Figure 12). That shows an iMessage-registered identifier on the device. It does not by itself show whose account it is. Accounts3.sqlite, com.apple.mobile.ldbbackup.plist and com.apple.locationd.plist were not captured, so iCloud account evidence, backup state, Find My iPhone state and the automatic-time setting remain open.

4.3 Settings and restrictions

PublicEffectiveUserSettings.plist (Figure 4) contains a "union" dictionary. Its removedSystemAppBundleIDs array lists five built-in apps: Shortcuts, Books, TV, Tips and the iCloud Drive app. This indicates that the user, or a configuration, removed these default apps. The visible restrictedBool entries are default-style values (for example forceGrantSupervisorAccessInAppLock is false). The exhibit contains no management or supervision indicators, but the whole file was not reviewed.

4.4 Applications and home-screen layout

SpringBoard IconState.plist describes the home-screen layout. The dock (buttonBar) holds Phone, Messages, Safari and Mail (Figure 8). A folder named Health holds com.apple.Health, com.apple.Fitness, com.fitbit.FitbitMobile and com.redshiftdev.Fitbit-Health-Sync (Figure 5). Two consecutive arrays list 37 further bundle identifiers (Figure 6 and Figure 7). They include messaging and calling apps (imo, LINE, Signal, Telegram, WhatsApp, Zoom and others), social platforms, secure e-mail, a Tor-based browser, vault applications and one identifier, kjc.loader, not resolved to a known application.

Two cautions apply. First, a bundle identifier in IconState.plist records layout. It is not proof of installation, launch or use, because layouts can retain identifiers for apps that are later offloaded or removed. Second, an application bundle (under /var/containers/Bundle/Application) holds the program, while the user data and preferences sit in a separate sandbox container (under /var/mobile/Containers/Data/Application/<GUID>). The GUID-named data containers were listed in the file manager but were not mapped to bundle identifiers in these exhibits, so no application-specific preference is reported.

4.5 Jailbreak indicators

No jailbreak-related file, folder or configuration artefact was documented in the exhibits. This is not a finding of absence: the relevant locations (for example checkra1n.dmg and Cydia-related paths in Appendix A of the brief) were not searched in the material compiled here. Any indicator found should be reported as an observable file only; a single artefact would not prove the historical state of the device.

5 Artefact Findings

5.1 Messages (sms.db)

sms.db (/private/var/mobile/Library/SMS/sms.db) contains 16 tables (Figure 9). The chat table lists at least 17 chats (Figure 12): 16 on the SMS service, using five-digit short codes and full telephone numbers, and one iMessage chat. Every SMS chat shares the same account identifier and last-addressed handle, which is the device's own number.

The first seven message rows (Figure 13) are all service SMS with the device's own account and handle IDs 1 to 7. Their text is automated: verification codes or alerts from Google, imo, LINE, an unnamed app, Signal, a social platform and Telegram. Codes are masked below.

ROWID	Content (masked)	handle_id	Raw date (ns)	UTC (converted)
1	Google verification code (code masked)	1	606532259367563904	2020-03-22 01:10:59
2	imo verification code (masked)	2	606582976713033984	2020-03-22 15:16:16
3	LINE verification prompt (code masked)	3	606597700687360000	2020-03-22 19:21:40
4	App password-reset code (service not named in the visible text)	4	606618398374657024	2020-03-23 01:06:38
5	Signal verification code (masked)	5	606620283330855936	2020-03-23 01:38:03
6	Notification alert from a social platform (text begins with a username)	6	606678096281632000	2020-03-23 17:41:36
7	Telegram verification code (masked)	7	606705694232059008	2020-03-24 01:21:34

Table 1. sms.db message rows 1 to 7. Conversion: raw ns / 10⁹ + 978,307,200 = Unix seconds. EDT is UTC minus 4 hours.

The message type column reads 0 for these rows, but the is_from_me and is_sent columns were not captured. Direction is therefore inferred from the content (service-originated codes), not read from the database, and is stated here as an inference. Whether any of these messages was acted on is not shown by sms.db.

The attachment table holds seven records (Figure 10 and Figure 11). Two are hidden plugin payloads (rich-message payloads with no MIME type), and five are visible files: a photo that appears twice (a staging copy with transfer_state 0 and a final copy with transfer_state 5), a second image, and two PDF documents. The is_outgoing flag shows which direction each attachment travelled.

ROWID	transfer_name	MIME	is_outgoing	state	Bytes
1	AF54E2D6-...(hidden plugin payload)	NULL	1	5	770,534

ROWID	transfer_name	MIME	is_outgoing	state	Bytes
2	IMG_0008.JPG (staged in /var/tmp/com.apple.messages)	image/jpeg	1	0	676,542
3	IMG_0008.JPG (~/Library/SMS/Attachments)	image/jpeg	1	5	676,542
4	FullSizeRender.jpeg	image/jpeg	0	5	368,536
5	How was an iPhone set up?.pdf	application/pdf	1	5	846,528
6	iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf	application/pdf	0	5	7,073,72
7	069EBD0E-...(hidden plugin payload)	NULL	1	5	1,598

Table 2. sms.db attachment records. Raw created_date values: 607027332; 608389002; 608389004; 608389082; 608390156; 608390475; 608730349 (Cocoa seconds).

5.2 Contacts, call history and voicemail

AddressBook.sqlitedb contains 30 tables (Figure 14). The ABPerson table holds two records that were examined. ROWID 1 (name masked) was created at raw 606532293, which is 2020-03-22 01:11:33 UTC, only 34 seconds after the first message in Table 1, and last modified at raw 607027441 (2020-03-27 18:44:01 UTC). ROWID 2, stored as "This Is DFIR", was created at raw 607027393 (2020-03-27 18:43:13 UTC) and modified at raw 607027406 (18:43:26 UTC). The phone and e-mail values for these records live in ABMultiValue and were not examined, so no contact was matched to a number.

CallHistory.storedata, table ZCALLRECORD, shows 20 records in the visible range (Figure 15 and Figure 16), all on the com.apple.Telephony service. Nineteen are from one San Jose, CA number (masked) with duration 0.0 seconds, ZANSWERED 0 and ZDISCONNECTED_CAUSE 6, at one to three attempts per day between 2020-03-23 20:02:52 UTC and 2020-04-09 15:54:57 UTC. The 20th record, raw ZDATE 607812138.04 (2020-04-05 20:42:18 UTC), is from a Fuquay-Varina, NC number (masked) with duration 23.09 seconds and a different disconnect cause (0).

These records show a repeated pattern of zero-duration contacts and one connected-length call. The ZLOCATION text is a region derived from the number, not the position of the handset or caller. Call direction (ZORIGINATED) was not captured. The 23-second record also reads ZANSWERED 0, which sits awkwardly beside a non-zero duration, so it is reported as an unresolved conflict rather than as an answered call. Section 7 covers this.

voicemail.db contains the tables voicemail, sqlite_sequence and _SqliteDatabaseProperties, with two indices (Figure 17). The voicemail table returns no rows in the Browse Data view. This is a supported finding that no voicemail metadata is recorded in this database. It does not prove that no voicemail audio exists, because the Voicemail folder itself was not reviewed.

Correlation attempt. No saved contact, call number or message sender could be matched across these three sources with the exhibits available. A saved contact would in any case not prove who used a number at a given time. The cross-source relationships that can be shown are in Section 6.

5.3 Calendar and organiser

Calendar.sqlitedb contains 38 tables including CalendarItem, Participant, Alarm, Location and Notification (Figure 18). Extras.db and several Notifications.Calendar files sit in the same folder. The exhibits show structure only, not row counts or events, so neither events nor an empty calendar is reported.

5.4 Safari history, bookmarks, tabs and cloud tabs

History.db has ten tables. The history_visits table (Figure 19) records a continuous browsing session on the evening of 27 March 2020 (EDT), shown below in UTC. It begins with searches about the 2020 baseball and hockey seasons, moves through news pages and Apple pages, and ends on two DFIR Review articles. Paired visit rows a second or so apart for the same item (for example ids 1 and 2) are typical of page loads that redirect or reload and should not be counted as separate user actions.

UTC (converted)	Raw visit_time	Title	Table
2020-03-28 00:58:37	607049917.431689	when does mlb start 2020 - Google Search	history_visits id 1
2020-03-28 00:59:03	607049943.212165	MLB 2020 season delayed	history_visits id 3
2020-03-28 01:02:44	607050164.953562	Is the NHL going to resume? - Google Search	history_visits id 8
2020-03-28 01:06:32	607050392.585617	Apple	history_visits id 12
2020-03-28 01:38:48	607052328.922416	Cult of Mac Tech and culture through an Apple lens	history_visits id 14
2020-03-28 01:44:37	607052677.913083	iOS Mobile Installation Logs - DFIR Review	history_visits id 16

Table 3. Selected Safari visits. EDT equivalents are 20:58 to 21:44 on 27 March 2020.

Bookmarks.db has six tables; its bookmarks bar holds the standard entries (iPhone User Guide, Apple, Bing, Google, Yahoo, MacRumors, Ars Technica, Cult of Mac), with no custom bookmark seen. BrowserState.db tabs (Figure 20) lists open tabs with last_viewed_time. Two of them match the History.db session: the NHL search tab (last viewed 607050365.05, 2020-03-28 01:06:05 UTC) and the DFIR Review tab (last_viewed_time -1.0, a non-time sentinel).

CloudTabs.db (Figure 21) holds exactly these two titles as synchronised tabs from a single device UUID. Later tab rows show a Fitbit e-mail verification URL (2020-04-03 18:51:36 UTC), a Fitbit login page (2020-04-06 01:15:20 UTC) and, at 2020-04-16 13:35:56 UTC, a Zoom sign-in page and a CoverMe confirmation URL recorded within a fraction of a millisecond of each other. All three services also appear among the application identifiers in Section 4.4.

The tabs table has a private_browsing column. The exhibit does not allow it to be read reliably for every row, so no private-browsing conclusion is drawn. History.db may also omit pages the user cleared, and last_viewed_time records the last view, not the first.

5.5 Photos and media

DCIM/100APPLE holds at least 59 numbered items (IMG_0001 to IMG_0059 in the visible grid), mixing JPG, PNG and MOV files (Figure 22). Photos.sqlite has 67 tables. In ZGENERICASSET (Figure 23) the ZADDEDDATE values include 608320569.66 (2020-04-11 17:56:09 UTC) and a group from 608394190.75 to 608395687.84 (2020-04-12 14:23:10 to 14:48:07 UTC). In the first 25 rows, four assets carry latitude and longitude (coordinates withheld here and redacted in the figure) and the others hold the sentinel -180.0, meaning no location. ZTRASHEDDATE is populated for rows 4 to 8, between 608395533.54 and 608395745.92 (2020-04-12 14:45:33 to 14:49:05 UTC), which marks those assets as moved to the trash. ZLASTSHAREDDATE is populated for several later rows (for example row 608745343, 2020-04-16 15:55:43 UTC).

ZADDITIONALASSETATTRIBUTES gives original file sizes, the time zone name America/New_York on every visible row, original file names that differ from the DCIM sequence (for example IMG_0727.jpg), EXIF time strings 2020:03:22 09:31:03 and 09:40:14 (text with no stated zone), and a creator bundle identifier of com.facebook.Messenger on one row. This metadata is an artefact claim: a GPS tag does not show that a named person took the picture.

Cross-artefact link. The attachment IMG_0008.JPG in sms.db is 676,542 bytes (Table 2). An asset in ZADDITIONALASSETATTRIBUTES row 8 has ZORIGINALFILESIZE 676,542 and the original file name IMG_0622.jpg. The identical byte count suggests the same image was received through Messages and saved to the library, with the file renamed during transfer. This is a size match only. A file hash comparison was not made, and the row alignment between the two Photos tables should be confirmed by Z_PK before the link is relied on.

5.6 KnowledgeC

The file /private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db was not opened in the exhibits compiled for this draft, so no ZOBJECT stream is reported and the chronology does not rely on it.

5.7 Health

The Health folder holds healthdb.sqlite and healthdb_secure.sqlite with their -shm and -wal files, plus Client, CloudSyncJournals, Journals and NanoSync folders. As a general rule, healthdb_secure.sqlite holds the protected sample tables such as activity_caches, while healthdb.sqlite holds a smaller set of unprotected data. The activity_caches query in Figure 28 names healthdb_secure.sqlite.

The sources table (Figure 25) lists six data contributors: the iPhone, the Health app, three Apple Watch entries and the Sync Solver app (com.redshiftdev.Fitbit-Health-Sync). That app also appears in the Health folder of the home screen, which links the Health data to the app inventory. activity_caches (Figure 26) holds daily summaries from 2020-03-21 onward, for example 28.97 kcal and 54 steps on 21 March and 454.2 kcal and 5,288 steps on 22 March. The workouts join (Figure 27) shows six workouts of activity_type 37 between 2020-03-27 and 2020-04-03 (examiner-local), the first running 14:57:39 to 15:15:39 on 27 March.

Health data is used here only as limited corroboration that the handset and paired devices were active on given days. Contributor and device records show which devices wrote data, but they do not identify the wearer, and no medical or lifestyle inference is drawn.

5.8 Location

locationd holds cache_encryptedB.db with 34 tables, including CellLocation, LteCellLocation and MicroLocation tables (Figure 29). Routined Local.sqlite and Cache.sqlite are present in com.apple.routined. The exhibits show structure only, so no Wi-Fi, LTE or Frequent Location points are reported and no map was produced. The only location-like values documented are the four GPS-tagged photo assets (5.5), which are metadata claims, and the call-number regions (5.2), which describe numbers rather than the handset. Neither is treated as a device or person location

6 Event Chronology

Window and justification. The chosen window runs from 2020-03-27 00:00 UTC to 2020-03-28 12:00 UTC (36 hours; 2020-03-26 20:00 to 2020-03-28 08:00 EDT). It was chosen because it holds the densest run of records from five artefact families (Health, call history, messages, contacts and Safari) within a single day and evening. The time-zone basis is America/New_York (UTC minus 4 hours on these dates) from Photos metadata. Raw values are Cocoa seconds unless stated.

#	UTC (converted)	EDT	Raw value	Source / table	Event	Confidence
E1	2020-03-27 14:57:39 to 15:15:39 (examiner-local)	Not converted	Not captured	Health: workouts join samples (data_id 15017, activity_type 37)	Workout record about 18 minutes long	Low to moderate: time zone of the examiner VM unverified
E2	2020-03-27 15:06:14 (examiner-local)	Not converted	Not captured	Health: source_devices	AirPods (0x2002) recorded as a data source	Low to moderate: same zone caveat; records the database entry, not pairing
E3	2020-03-27 16:25:36	12:25:36	607019136.714309	CallHistory.storedata / ZCALLRECORD	Zero-duration, unanswered record from one San Jose, CA number (masked)	High for time; direction not captured
E4	2020-03-27 18:42:12	14:42:12	607027332	sms.db / attachment ROWID 1	Hidden plugin-payload attachment (770,534 bytes) created, is_outgoing 1	High for time; content not recovered
E5	2020-03-27 18:43:13	14:43:13	607027393	AddressBook.sqlitedb / ABPerson ROWID 2	Contact record "This Is DFIR" created	High for time; creator unknown
E6	2020-03-27 18:43:26	14:43:26	607027406	AddressBook.sqlitedb / ABPerson ROWID 2	Same contact record modified	High for time
E7	2020-03-27 18:44:01	14:44:01	607027441	AddressBook.sqlitedb / ABPerson ROWID 1	Contact record ROWID 1 modified	High for time
E8	2020-03-27 19:55:03	15:55:03	607031703.669631	CallHistory.storedata / ZCALLRECORD	Second zero-duration, unanswered record from the same number	High for time; direction not captured
E9	2020-03-28 00:58:37	2020-03-27 20:58:37	607049917.431689	Safari History.db / history_visits id 1	Google search "when does mlb start 2020"	High for time; page load, not proof of typing

#	UTC (converted)	EDT	Raw value	Source / table	Event	Confidence
E10	2020-03-28 01:02:44	2020-03-27 21:02:44	607050164.953562	Safari History.db / history_visits id 8	Google search "Is the NHL going to resume?"	High; corroborated by E11
E11	2020-03-28 01:06:05	2020-03-27 21:06:05	607050365.052293	Safari BrowserState.db / tabs id 1; CloudTabs.db / cloud_tabs	Same NHL page held as an open tab and as a synchronised cloud tab	High; three independent Safari databases agree
E12	2020-03-28 01:44:37	2020-03-27 21:44:37	607052677.913083	Safari History.db / history_visits id 16; BrowserState.db / tabs id 2; CloudTabs.db	DFIR Review article "iOS Mobile Installation Logs" visited, then held as a tab and cloud tab	High; three independent Safari databases agree
E13	2020-03-28 01:46:11	2020-03-27 21:46:11	607052771.353487	Safari BrowserState.db / tabs id 5	Ars Technica tab last viewed	High for time; last view only

Table 4. Event chronology, 27 to 28 March 2020. Health rows E1 and E2 are in examiner-local time because the queries used the localtime modifier.

Corroborated and related events

Ref	Relationship	Independent sources	Strength and qualification
R1	NHL search page: visited at 01:02:44 UTC, open as tab 1, and synchronised as a cloud tab (E10, E11)	History.db, BrowserState.db, CloudTabs.db	Strong: same title and URL stem in three databases. Shows the page was open on the device and synced; does not show who was viewing.
R2	DFIR Review article: visited at 01:44:37 UTC, open as tab 2, and synchronised as a cloud tab (E12)	History.db, BrowserState.db, CloudTabs.db	Strong for the same reasons as R1.
R3	Hidden attachment created at 18:42:12 UTC; contact record "This Is DFIR" created 61 seconds later at 18:43:13 UTC (E4, E5)	sms.db, AddressBook.sqlitedb	Moderate to low: close timing across independent databases, but the attachment has no MIME type, so a contact share is possible and unproven.
R4	Outside the window: attachment IMG_0008.JPG (676,542 bytes) matches a Photos asset of the same size (Section 5.5)	sms.db, Photos.sqlite	Moderate: byte-size match only; hash and Z_PK alignment not confirmed.
R5	Message codes from imo, LINE, Signal and Telegram (Table 1) and the same apps in IconState.plist (Section 4.4)	sms.db, IconState.plist	Moderate: consistent, but a verification SMS does not prove the app was installed or used.

7 Interpretation and Limits

7.1 What the evidence supports

- The Health source records show an iPhone8,4 that ran iOS 13.3.1 and 13.4.1, with an Apple Watch (Watch4,3), AirPods and AirPods Pro as additional data sources.
- sms.db holds automated verification and alert messages and five visible attachments with direction and size recorded; Safari records a connected evening session confirmed in three databases (Section 6).

7.2 What remains uncertain

- Serial number, ICCID, IMSI, MCC, MNC, iCloud account state, backup state, Find My iPhone state and the automatic-time setting are not established.
- KnowledgeC streams, Calendar rows and any location rows have not been examined, so no finding depends on them and no absence is claimed.
- The direction of calls and messages, and whether any message was read or acted on, is not read from the databases.
- No contact was matched to a call number or message sender, and a saved contact would not by itself prove who used a number.

7.3 Contradictions and alternative explanations

- Call record 31 shows ZANSWERED 0 together with a 23.09 second duration and a disconnect cause of 0, unlike the other records (cause 6). The duration suggests a connected call, but the flag says not answered. Either the flag is not a reliable indicator for this call type or the duration includes ringing time; neither can be settled from the exhibit.
- Photo IMG_0008.JPG has a different original file name (IMG_0622.jpg). Messages renames received images, which explains the difference, but the explanation depends on the row alignment noted in Section 5.5.
- A device-level artefact does not identify a person. A session on the phone could have been a different user, an automated process or a synchronised copy from another device. CloudTabs.db, for example, exists to reflect tabs from synchronised devices.

7.4 Four evidence-based findings

Finding 1: device and account context. The handset was an iPhone8,4 that ran iOS 13.3.1 and 13.4.1, was used with an Apple Watch and two AirPods models, has a US-numbered SMS account and a separate iMessage identifier, and shows America/New_York time-zone metadata. Limit: the subscriber, SIM and iCloud identifiers are still to be confirmed from the plists in Appendix D.

Finding 2: communications and organiser activity. Messages are dominated by automated verification codes from messaging services. A repeated zero-duration call pattern and one 23-second call are recorded, and the voicemail table is empty. Limit: direction was not captured, no contact was correlated, and calendar content was not reviewed.

Finding 3: web, media and application activity. Safari shows a connected evening session on 27 March 2020 (EDT), later Fitbit, Zoom and CoverMe pages, and matching app identifiers in the home-screen layout. Photos.sqlite records at least 59 DCIM items, with add, share and trash dates and a few GPS tags. Limit: bundle identifiers in IconState.plist show layout and not use, and photo metadata is an artefact claim.

Finding 4: reconstructed chronology. Thirteen events from five artefact families fit a 36-hour window, with two Safari relationships corroborated across three databases and three further related events. Limit: KnowledgeC is missing, two Health rows are in examiner-local time, and relationship R3 is circumstantial.

Closing statement. This report describes what the artefacts record. It does not decide whether any person committed an offence, and it does not identify the person who used the device at any time.

Appendix A Finding Log

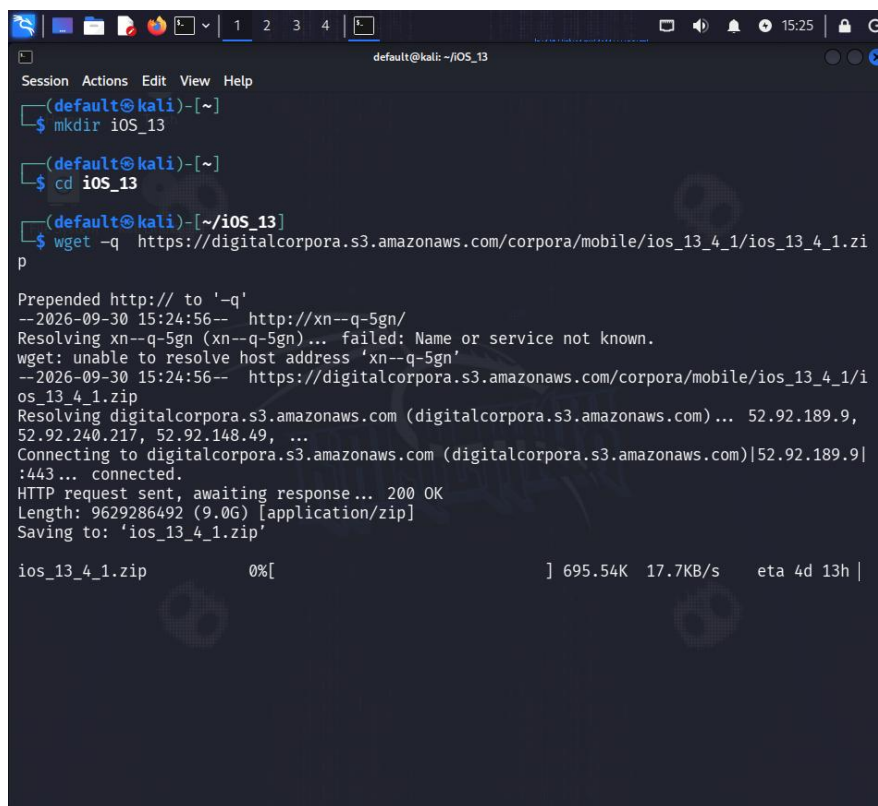
The format-example row in the brief has been removed. Every row retains source, raw value, converted time and qualification.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-01	Package integrity: MD5 identical to brief; SHA-256 identical to the 64-character workstation value; brief value is 63 characters	ios_13_4_1.zip (hashdeep)	Size 9,629,286,492; MD5 4e09...0d1d	Not applicable	Likely typo in brief; confirm with instructor
F-02	File-system tree extracted to Volumes/JOSH/NoTar-13-4-1/	13-4-1.tar (tar -xvf)	Not applicable	Examiner time 30 Sep 2026	Extraction listing only; original ZIP untouched
F-03	iPhone8,4 recorded at iOS 13.3.1 and 13.4.1	Health DB / source_devices	creation_date not captured	2020-03-21 18:33:20; 2020-04-13 15:26:44 (examiner-local)	localtime modifier; database file not shown
F-04	Apple Watch Watch4,3 6.1.3 to 6.2.1; AirPods; AirPods Pro	Health DB / source_devices	creation_date not captured	2020-03-21 16:28:48; 2020-03-27 15:06:14; 2020-03-31 15:08:08; 2020-04-13 21:16:00 (examiner-local)	Records database entry, not pairing time
F-05	Five built-in apps removed (Shortcuts, Books, TV, Tips, iCloud Drive)	PublicEffectiveUserSettings.plist / union / removedSystemAppBundleIDs	Array of 5 strings	Not applicable	Does not show who removed them or when
F-06	Dock apps; Health folder; 37 further bundle identifiers	SpringBoard/IconState.plist	iconLists, buttonBar	Not applicable	Layout, not proof of installation or use
F-07	sms.db: 16 tables, 7 attachments, 17+ chats (16 SMS, 1 iMessage)	/private/var/mobile/Library/SMS/sms.db	Not applicable	Not applicable	Direction columns not captured
F-08	Messages ROWID 1 to 7: automated verification codes and one alert	sms.db / message	606532259367563904 to 606705694232059008 (ns)	2020-03-22 01:10:59 to 2020-03-24 01:21:34 UTC	Codes masked; direction inferred
F-09	Attachments: IMG_0008.JPG (x2), FullSizeRender.jpeg, 2 PDFs, 2 hidden payloads	sms.db / attachment	607027332 to 608730349	2020-03-27 18:42:12 to 2020-04-16 11:45:49 UTC	Content of hidden payloads not recovered
F-10	Contact records ROWID 1 and 2 created and modified	AddressBook.sqlitedb / ABPerson	606532293; 607027393; 607027406; 607027441	2020-03-22 01:11:33; 2020-03-27 18:43:13; 18:43:26; 18:44:01 UTC	Numbers not matched; ABMultiValue not examined
F-11	20 call records: 19 zero-duration from one number; 1 of 23.09 s from another	CallHistory.storedata / ZCALLRECORD	606686572.41 to 608140497.05; 607812138.04 (23.09 s)	2020-03-23 20:02:52 to 2020-04-09 15:54:57 UTC; 2020-04-05 20:42:18 UTC	ZANSWERED 0 conflicts with 23.09 s; direction not captured
F-12	No voicemail rows	voicemail.db / voicemail	Not applicable	Not applicable	Audio files not reviewed
F-13	Safari session about sports news, Apple and DFIR Review pages	Safari/History.db / history_visits	607049917.43 to 607052677.91	2020-03-28 00:58:37 to 01:44:37 UTC	Redirect pairs should not be double-counted
F-14	NHL and DFIR Review tabs open and synchronised	BrowserState.db / tabs; CloudTabs.db / cloud_tabs	607050365.05 (tab 1); -1.0 (tab 2)	2020-03-28 01:06:05 UTC (tab 1)	Tab 2 sentinel; private_browsing not read
F-15	Fitbit, Zoom and CoverMe pages in tabs	BrowserState.db / tabs	607632696.37; 607828520.50; 608736956.157	2020-04-03 18:51:36; 2020-04-06 01:15:20; 2020-04-16 13:35:56 UTC	Last view only

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-16	DCIM/100APPLE 59+ items; asset add, trash and share dates; 4 GPS-tagged of first 25	Photos.sqlite / ZGENERICASSET; /var/mobile/Media/DCIM	608320569.66; 608394190.75; 608395533.54	2020-04-11 17:56:09; 2020-04-12 14:23:10; 2020-04-12 14:45:33 UTC	Metadata is an artefact claim; coordinates withheld
F-17	IMG_0008.JPG attachment size matches a Photos asset (676,542 bytes)	sms.db / attachment; Photos.sqlite / ZADDITIONALASSETATTRIBUTES	608389004 (attachment created)	2020-04-12 12:56:44 UTC	Size match only; hash not compared
F-18	Daily activity summaries and six workouts of type 37	Health DB / activity_caches; workouts	cache_index values; workout times not captured	Workout 1: 2020-03-27 14:57:39 to 15:15:39 (examiner-local)	Limited corroboration; no medical inference
F-19	cache_encryptedB.db schema (34 tables); routined files present	var/root/Library/Caches/locationd; var/mobile/Library/Caches/com.apple.routined	Not applicable	Not applicable	Structure only; no rows reviewed
F-20	Calendar.sqlitedb schema (38 tables); Extras.db present	var/mobile/Library/Calendar	Not applicable	Not applicable	Structure only; no rows reviewed

Appendix B Screenshots with Captions

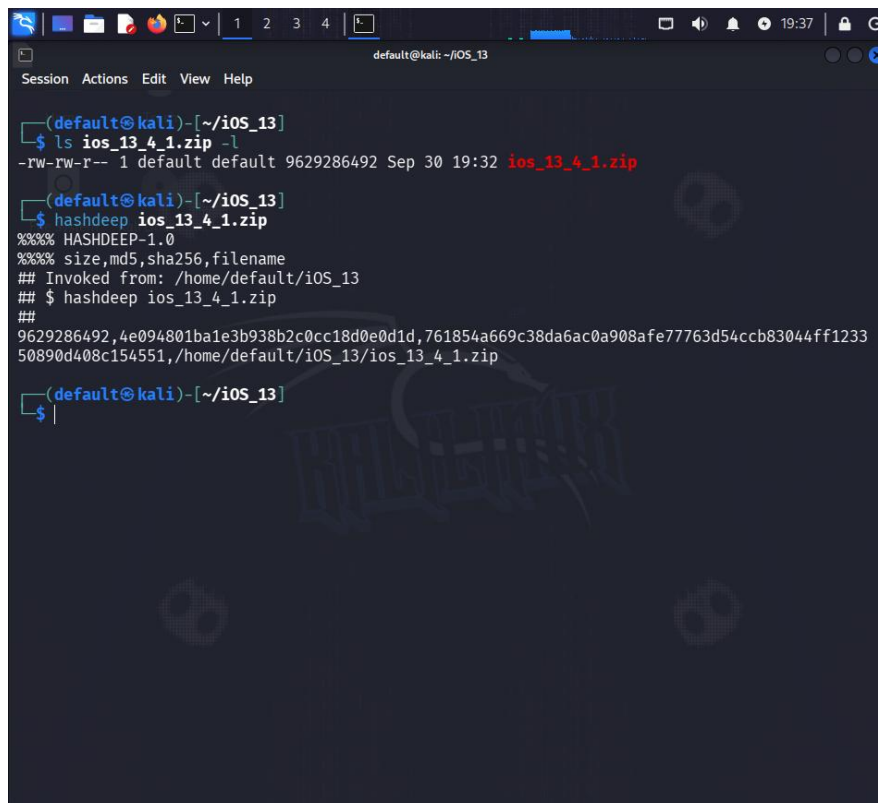
Figures are the examiner's own screenshots from the Kali workstation. Each caption names the source, table or file shown. Telephone numbers, e-mail-format accounts and GPS coordinates are redacted in Figures 12, 13, 16, 23.



```
default@kali: ~/iOS_13
Session Actions Edit View Help
(default@kali)-[~]
$ mkdir iOS_13
(default@kali)-[~]
$ cd iOS_13
(default@kali)-[~/iOS_13]
$ wget -q https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip
p
Prepended http:// to '-q'
--2026-09-30 15:24:56-- http://xn--q-5gn/
Resolving xn--q-5gn (xn--q-5gn)... failed: Name or service not known.
wget: unable to resolve host address 'xn--q-5gn'
--2026-09-30 15:24:56-- https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/i
os_13_4_1.zip
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 52.92.189.9,
52.92.240.217, 52.92.148.49, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.189.9|
:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9629286492 (9.0G) [application/zip]
Saving to: 'ios_13_4_1.zip'

ios_13_4_1.zip      0%[                ] 695.54K  17.7KB/s   eta 4d 13h |
```

Figure 1. Retrieval of the training image with wget into ~/iOS_13 on the Kali examination workstation (30 Sep 2026). The first command contained a mistyped option and returned a harmless name-resolution error; the download then started from the Digital Corpora source URL (length 9,629,286,492 bytes). Output is truncated at 0%; completion is evidenced by Figure 2.

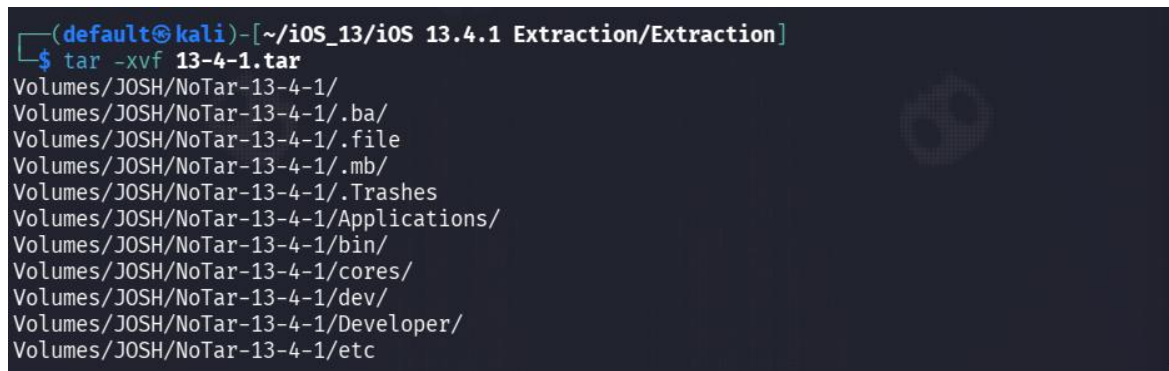


```
(default@kali)~/iOS_13
$ ls ios_13_4_1.zip -l
-rw-rw-r-- 1 default default 9629286492 Sep 30 19:32 ios_13_4_1.zip

(default@kali)~/iOS_13
$ hashdeep ios_13_4_1.zip
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
## Invoked from: /home/default/iOS_13
## $ hashdeep ios_13_4_1.zip
##
9629286492,4e094801ba1e3b938b2c0cc18d0e0d1d,761854a669c38da6ac0a908afe77763d54ccb83044ff1233
50890d408c154551,/home/default/iOS_13/ios_13_4_1.zip

(default@kali)~/iOS_13
$
```

Figure 2. *ls* and *hashdeep* output for *ios_13_4_1.zip*: size 9,629,286,492 bytes, MD5 and SHA-256 values used in the integrity check (Section 3.2).



```
(default@kali)~/iOS_13/iOS 13.4.1 Extraction/Extraction
$ tar -xvf 13-4-1.tar
Volumes/JOSH/NoTar-13-4-1/
Volumes/JOSH/NoTar-13-4-1/.ba/
Volumes/JOSH/NoTar-13-4-1/.file
Volumes/JOSH/NoTar-13-4-1/.mb/
Volumes/JOSH/NoTar-13-4-1/.Trashes
Volumes/JOSH/NoTar-13-4-1/Applications/
Volumes/JOSH/NoTar-13-4-1/bin/
Volumes/JOSH/NoTar-13-4-1/cores/
Volumes/JOSH/NoTar-13-4-1/dev/
Volumes/JOSH/NoTar-13-4-1/Developer/
Volumes/JOSH/NoTar-13-4-1/etc
```

Figure 3. *Extraction of 13-4-1.tar with tar -xvf; the file-system tree is written under Volumes/JOSH/NoTar-13-4-1/. The original ZIP was not modified.*

```
(kali@kali)-[~/mobile/Library/UserConfigurationProfiles/PublicInfo]
$ plistutil -i PublicEffectiveUserSettings.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>union</key>
  <dict>
    <key>audioCallDefaultHandlerAppBundleIDs</key>
    <dict/>
    <key>knownAirPrintTargets</key>
    <dict/>
    <key>tvRemoteAllowedRemotes</key>
    <dict/>
    <key>managedWebDomains</key>
    <dict/>
    <key>trustedCodeSigningIdentities</key>
    <dict/>
    <key>removedSystemAppBundleIDs</key>
    <dict>
      <key>values</key>
      <array>
        <string>com.apple.shortcuts</string>
        <string>com.apple.iBooks</string>
        <string>com.apple.tv</string>
        <string>com.apple.tips</string>
        <string>com.apple.iCloudDriveApp</string>
      </array>
    </dict>
    <key>tvRemoteAllowedTVs</key>
    <dict/>
    <key>allowedSafariPasswordAutoFillDomains</key>
    <dict/>
    <key>webContentFilterBlacklistedURLs</key>
    <dict/>
  </dict>
  <key>restrictedBool</key>
  <dict>
    <key>forceGrantSupervisorAccessInAppLock</key>
    <dict>
      <key>value</key>
      <false/>
    </dict>
  </dict>
</plist>
```

Figure 4. *plistutil* rendering of `/var/mobile/Library/UserConfigurationProfiles/PublicInfo/PublicEffectiveUserSettings.plist`; the `removedSystemAppBundleIDs` array lists five removed built-in apps.

```
<key>listUniqueIdentifiers</key>
<array>
  <string>8AEA3798-8CC0-42A9-8C99-E2934FBE52C2</string>
</array>
<key>iconLists</key>
<array>
  <array>
    <string>com.apple.Health</string>
    <string>com.apple.Fitness</string>
    <string>com.fitbit.FitbitMobile</string>
    <string>com.redshiftdev.Fitbit-Health-Sync</string>
  </array>
</array>
<key>listType</key>
<string>folder</string>
<key>displayName</key>
<string>Health</string>
<key>uniqueIdentifier</key>
<string>8A78AA4A-A95A-4505-A5FD-9E888F1C80FE</string>
```

Figure 5. *SpringBoard IconState.plist*: folder "Health" holding `com.apple.Health`, `com.apple.Fitness`, `com.fitbit.FitbitMobile` and `com.redshiftdev.Fitbit-Health-Sync`.

```
<array>
  <string>com.hammerandchisel.discord</string>
  <string>com.mentionmobile.cyberdust</string>
  <string>com.facebook.Messenger</string>
  <string>org.mozilla.ios.Focus</string>
  <string>imgurmobile</string>
  <string>co.babypenguin.imo</string>
  <string>com.burbn.instagram</string>
  <string>com.kik.chat</string>
  <string>jp.naver.line</string>
  <string>com.mewe</string>
  <string>com.reddit.Reddit</string>
  <string>org.whispersystems.signal</string>
  <string>com.silentcircle.SilentPhone</string>
  <string>com.skout.SKOUT</string>
  <string>com.skype.skype</string>
  <string>com.toyopagroup.picaboo</string>
  <string>com.spotify.client</string>
  <string>ph.telegra.Telegraph</string>
  <string>com.tinginteractive.usms</string>
  <string>com.burbn.threads</string>
</array>
```

Figure 6. SpringBoard IconState.plist: first array of 20 messaging, social, privacy and utility bundle identifiers (Discord, imo, LINE, Signal, Telegram and others).

```
<array>
  <string>com.zhiliaoapp.musically</string>
  <string>com.miketigas.OnionBrowser</string>
  <string>com.atebits.Tweetie2</string>
  <string>net.kortina.labs.Venmo</string>
  <string>com.viber</string>
  <string>net.whatsapp.WhatsApp</string>
  <string>com.mywickr.wickr</string>
  <string>com.wearezeta.zclient.ios</string>
  <string>ch.protonmail.protonmail</string>
  <string>de.tutao.tutanota</string>
  <string>com.adhoclabs.burner</string>
  <string>com.herzick.houseparty</string>
  <string>us.zoom.videomeetings</string>
  <string>kjc.loader</string>
  <string>com.coverme.covermeAdhoc</string>
  <string>com.keepsafe.KeepSafe</string>
  <string>com.enchantedcloud.photovault</string>
</array>
```

Figure 7. SpringBoard IconState.plist: continuation array of 17 bundle identifiers (WhatsApp, Zoom, KeepSafe, CoverMe, Photo Vault and others).

```

<key>buttonBar</key>
<array>
    <string>com.apple.mobilephone</string>
    <string>com.apple.MobileSMS</string>
    <string>com.apple.mobilesafari</string>
    <string>com.apple.mobilemail</string>
</array>
<key>displayName</key>
<string>Folder</string>
<key>defaultDisplayName</key>
<string>Folder</string>
<key>uniqueIdentifier</key>
<string>7985B55E-CE7E-4317-A69C-252A0DEBFB0E</string>

```

Figure 8. IconState.plist buttonBar (dock): Phone, Messages, Safari and Mail.

Name	Type	Schema
Tables (16)		
_SqliteDatabaseProperties		CREATE TABLE _SqliteDatabaseProperties (key TEXT, value TEXT, UNIQUE(key))
attachment		CREATE TABLE attachment (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT U
chat		CREATE TABLE chat (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT U
chat_handle_join		CREATE TABLE chat_handle_join (chat_id INTEGER REFERENCES chat (ROWID) ON D
chat_message_join		CREATE TABLE chat_message_join (chat_id INTEGER REFERENCES chat (ROWID) ON
deleted_messages		CREATE TABLE deleted_messages (ROWID INTEGER PRIMARY KEY AUTOINCREMENT
handle		CREATE TABLE handle (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, i
kvtable		CREATE TABLE kvtable (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, i
message		CREATE TABLE message (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TE
message_attachment_join		CREATE TABLE message_attachment_join (message_id INTEGER REFERENCES messa
message_processing_task		CREATE TABLE message_processing_task (ROWID INTEGER PRIMARY KEY AUTOINC
sqlite_sequence		CREATE TABLE sqlite_sequence(name,seq)
sqlite_stat1		CREATE TABLE sqlite_stat1(tbl,idx,stat)
sync_deleted_attachments		CREATE TABLE sync_deleted_attachments (ROWID INTEGER PRIMARY KEY AUTOINC
sync_deleted_chats		CREATE TABLE sync_deleted_chats (ROWID INTEGER PRIMARY KEY AUTOINCREM
sync_deleted_messages		CREATE TABLE sync_deleted_messages (ROWID INTEGER PRIMARY KEY AUTOINC

Figure 9. sms.db opened in DB Browser for SQLite: 16 tables including attachment, chat, handle, message and the deleted/sync_deleted tables.

ROWID	guid	created_date	start_date	filename
Filter	Filter	Filter	Filter	Filter
1	at_0_57633602-7DC8-4506-AA44-9F06D82CA654	607027332	0	~/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-E9FC69565B4D/AF54E2D6-D6C5-4652-ABED-58C6D0662D0E.pluginPayloadAttachment
2	6B32DD57-049C-405B-8A4A-CAE8B5C201E1	608389002	0	/var/tmp/com.apple.messages/com.apple.MobileSMS/Media/ES4EFEAC-40A3-45D3-88BD-84B3E0A576FC/IMG_0008.JPG
3	at_0_784979CA-1E85-4FFE-A2CE-1AD0B388209E	608389004	0	~/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638B8CF97B/IMG_0008.JPG
4	at_0_C8BCB97D-9503-48BD-842D-7618ADD1631E	608389082	608389082	~/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-ABD7-218D94989A09/FullSizeRender.jpeg
5	at_0_D3CBA063-6CEB-4326-84BB-10D68D9F2E50	608390156	0	~/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/How was an iPhone set up?.pdf
6	at_0_6A995D6E-14E2-4620-B826-8F4EE2926FE5	608390475	608390476	~/Library/SMS/Attachments/97/07/476F18F4-4FE5-466A-AD8C-605576C5F16D/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf
7	at_0_E1DE4D6D-5E22-44CB-87AA-A173AAA77DE0	608730349	0	~/Library/SMS/Attachments/db/11/0DDBCB8D-6270-4980-85B1-ADC00A191D32/069EBD0E-FDA6-486A-AA02-49E389691551.pluginPayloadAttachment

Figure 10. sms.db attachment table (part 1): ROWID, guid, created_date, start_date and filename for seven attachment records.

	mime_type	transfer_state	is_outgoing	user_info	transfer_name	total_bytes	is_sticker	sticker_user_info	attribution_info	hide_attachment
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	NULL	5	1	NULL	AF54E2D6-D6C5-4652-...	770534	0	NULL	NULL	1
2	image/jpeg	0	1	NULL	IMG_0008.JPG	676542	0	NULL	NULL	0
3	image/jpeg	5	1	BLOB	IMG_0008.JPG	676542	0	NULL	BLOB	0
4	image/jpeg	5	0	BLOB	FullSizeRender.jpeg	368536	0	NULL	BLOB	0
5	application/pdf	5	1	BLOB	How was an iPhone set up?.pdf	846528	0	NULL	NULL	0
6	application/pdf	5	0	BLOB	iOS_Bug_Reporting_for_Forensic_Purposes...	7073725	0	NULL	NULL	0
7	NULL	5	1	NULL	069EBD0E-FDA6-486A-...	1598	0	NULL	NULL	1

Figure 11. sms.db attachment table (part 2): MIME type, transfer_state, is_outgoing, transfer_name and total_bytes. Source for Table 3.

Figure 12. sms.db chat table. Chat identifiers, GUIDs and account handles are redacted in this figure (telephone numbers and an e-mail-format account); unredacted originals remain in the examiner working copy.

Figure 13. sms.db message table (ROWID 1 to 7). The account and account_guid columns are redacted. The date column holds nanosecond Cocoa values.

Figure 14. *AddressBook.sqlitedb* schema listing 30 tables, including *ABPerson*, *ABMultiValue* and *ABPersonFullTextSearch*.

Table: ZCALLRECORD										
	_PK	Z_ENT	Z_OPT	ZANSWERED	ZCALL_CATEGORY	ZCALLTYPE	ZDISCONNECTED_CAUSE	ZFACE_TIME_DATA	ZFILTERED_OUT_REASON	
	Filter	Filter	Filter	Filter	Filter	Filter		Filter	Filter	
1	12	2	2	0	1	1	6	NULL		0
2	16	2	2	0	1	1	6	NULL		0
3	26	2	2	0	1	1	6	NULL		0
4	27	2	2	0	1	1	6	NULL		0
5	28	2	2	0	1	1	6	NULL		0
6	29	2	2	0	1	1	6	NULL		0
7	30	2	2	0	1	1	6	NULL		0
8	31	2	1	0	1	1	0	NULL		0
9	32	2	2	0	1	1	6	NULL		0
10	33	2	2	0	1	1	6	NULL		0
11	34	2	2	0	1	1	6	NULL		0
12	35	2	2	0	1	1	6	NULL		0
13	36	2	2	0	1	1	6	NULL		0
14	37	2	2	0	1	1	6	NULL		0
15	38	2	2	0	1	1	6	NULL		0
16	39	2	2	0	1	1	6	NULL		0
17	40	2	2	0	1	1	6	NULL		0
18	41	2	2	0	1	1	6	NULL		0

Figure 15. *CallHistory.storeddata*, table ZCALLRECORD: ZANSWERED, ZCALLTYPE and ZDISCONNECTED_CAUSE for the first 18 records.

ZDATE	ZDURATION	ZISO_COUNTRY_CODE	ZLOCATION	ZNAME	ZSERVICE_PROVIDER	QUANTITY	ADDRESS		
Filter	Filter	Filter	Filter	Filter	Filter	F...	F...	F...	Filter
606686572.413675		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
606764238.386421		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
606937905.317984		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607019136.714309		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607031703.669631		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607464398.417464		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607623054.597476		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607812138.039618	23.08800303936	us	Fuquay-Varina, NC	NULL	com.apple.Telephony	...	...	...	
607898073.986463		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607902488.087266		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607905780.616611		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607975842.40421		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607979094.489056		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
607983027.416206		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
608067126.706149		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
608068419.884882		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
608070119.134031		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
608076742.076481		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
608077065.571387		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	
608140497.052901		0.0 us	San Jose, CA	NULL	com.apple.Telephony	...	...	...	

Figure 16. ZCALLRECORD: ZDATE, ZDURATION, ZLOCATION and service provider. The ZADDRESS column is redacted in this figure.

Database Structure
Browse Data
Edit Pragmas
Execute SQL

Table: voicemail
Filter in any column

ROWID	remote_uid	date	token	sender	callback_num	duration	expiration	trashed_date	flags	receiver
Filter	Filter	Fi...	Filter	Filter	Filter	Filter	Filter	Filter	Fi...	Filter

Figure 17. *voicemail.db*, table *voicemail*: the Browse Data view returns no rows.

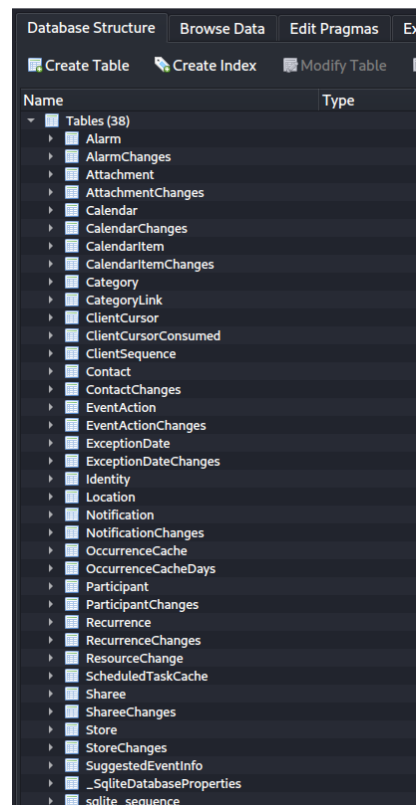


Figure 18. Calendar.sqlitedb opened in DB Browser for SQLite: 38 tables including CalendarItem, Participant, Alarm and Location (structure only).

Table: history_visits				Filter in any column	
	id	history_item	visit_time	title	
F...	Filter	Filter	Filter		
1	1	1	607049917.431689	when does mlb start 2020 - Google Search	
2	2	1	607049916.117145	when does mlb start 2020 - Google Search	
3	3	2	607049943.212165	MLB 2020 season delayed	
4	4	2	607049971.64417	MLB 2020 season delayed	
5	5	2	607049972.930589	MLB 2020 season delayed	
6	6	1	607050017.768145	when does mlb start 2020 - Google Search	
7	7	3	607050125.997712	when does mlb start 2020 - Google Search	
8	8	4	607050164.953562	Is the NHL going to resume? - Google Search	
9	9	4	607050164.282689	Is the NHL going to resume? - Google Search	
10	10	5	607050233.89337	NHL expects to resume season after pause for coronavirus	
11	11	5	607050231.330998	NHL expects to resume season after pause for coronavirus	
12	12	6	607050392.585617	Apple	
13	13	7	607050461.869823	iPad Pro - Apple	
14	14	8	607052328.922416	Cult of Mac Tech and culture through an Apple lens	
15	15	9	607052629.819015	DFIR Review	
16	16	10	607052677.913083	iOS Mobile Installation Logs - DFIR Review	

Figure 19. Safari History.db, table history_visits: raw visit_time values and page titles for the 28 March 2020 (UTC) browsing session.

Database Structure Browse Data Edit Pragas Execute SQL										
Table: tabs										
id	url	title	user_visible_url	order_index	last_viewed_time	..bo opened_from_link	ig_top	private_browsing	and win	browser_window_id
1	...	Is the NHL going to resume? - Google Search	https://www.google.com/search?...	0	607050365.052293	0	0	0	0	1
2	...	iOS Mobile Installation Logs - DFIR Review	https://dfir.pubpub.org/pub/e5xlbw88	1	-1.0	0	0	0	0	1
3	...	What Happened to Tiger King's Carole ...	https://www.google.com/amp/s/...	0	607051785.025869	0	0	0	0	1
4	...	Ars Technica	https://arstechnica.com/	1	607052771.353487	0	0	0	0	1
5	...	dfir.pubpub.org/pub/e5xlbw88	https://dfir.pubpub.org/pub/e5xlbw88	0	607052711.986211	0	1	0	0	2
6	...	How was an iPhone set up? - DFIR Review	https://dfir.pubpub.org/pub/2q177smo	2	607379019.952313	0	0	0	0	1
7	...	iOS - Home Accessories - Apple	https://www.apple.com/ios/home/...	3	607625374.024899	0	0	0	0	1
8	...	www.fitbit.com/verifyEmail/verify?...	https://www.fitbit.com/verifyEmail/verify?...	4	607632696.371496	0	0	0	0	1
9	...	Fitbit	https://accounts.fitbit.com/login?...	5	607828520.497337	0	0	0	0	1
10	...	Sign In - Zoom	https://zoom.us/signin	6	608736956.15698	0	0	0	0	1
11	...	http://s.coveme.ws/confirm?...	http://s.coveme.ws/confirm?...	7	608736956.156997	0	0	0	0	1

Figure 20. Safari BrowserState.db, table tabs: tab titles, URLs and last_viewed_time values.

Database StructureBrowse DataEdit PragasExecute SQL

Table: cloud_tabs

tab_uuid	system_fields	device_uuid	position	title	url
Filter	Filter	Filter	Filter	Filter	Filter
1 6DE6EA6D-6822-4623-A307-2805B949CC9F	BLOB	EB69B16A-E851-4D2A-96F1-9B154A2DEF39	BLOB	iOS Mobile Installation Logs - DFIR Review	https://dfir.pubpub.org/pub/e5xlbw88
2 40E8E562-38E1-4782-95F3-ACF3F7F166FE	BLOB	EB69B16A-E851-4D2A-96F1-9B154A2DEF39	BLOB	Is the NHL going to resume? - Google Search	https://www.google.com/search?...

Figure 21. Safari CloudTabs.db, table cloud_tabs: two synchronised tabs from one device UUID.

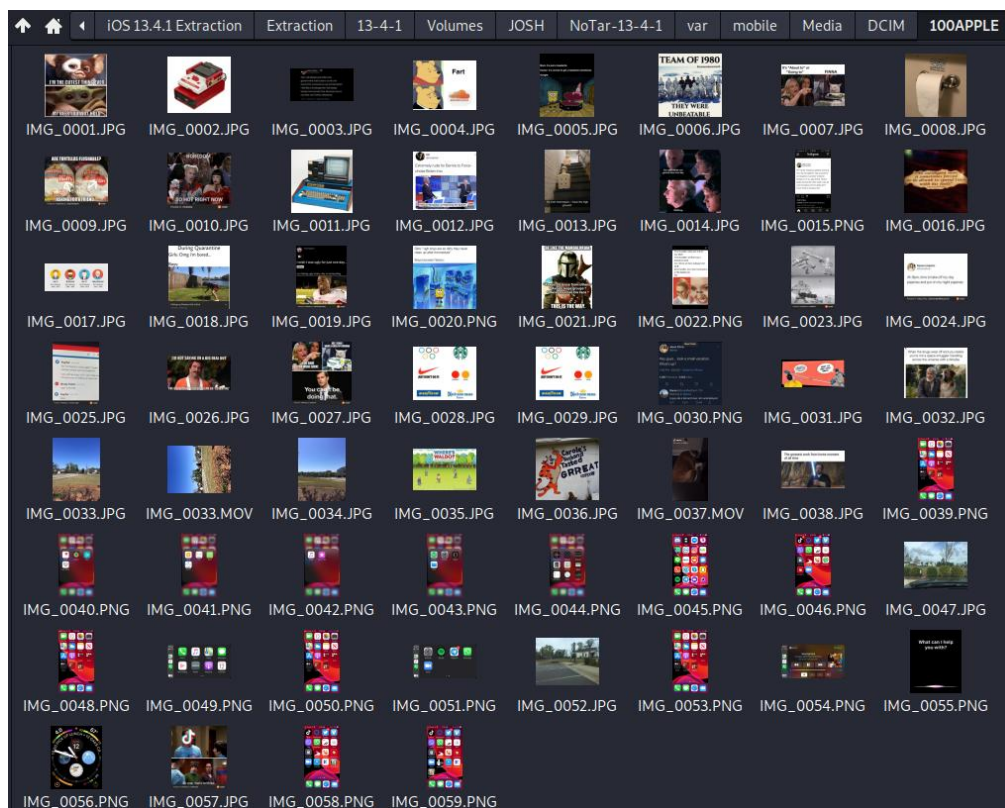


Figure 22. DCIM/100APPLE thumbnail view showing JPG, PNG and MOV items (IMG_0001 onward). Reduced scale; content not described in the narrative.

Database Structure																						Browse Data				Edit Pragmas				Execute SQL											
Table: ZGENERICASSET																																						Filter in any column			
	ZADDEDDATE	NT	MC	HF	WEI	ERI	TOI	CR	ZDURATION	TM	RG	ISI	ZLASTSHARED	DATE	ZLATITUDE	ZLONGITUDE	ZMODIFICATIONDATE	IST	TIC	TTI	ZTRASHEDDATE																				
Filter	F...	F...	F...	F...	F...	F...	F...	Filter	F...	F...	F...	Filter	Filter	Filter	Filter	Filter	Filter	F...	F...	F...	Filter																				
1	608320616.052433	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...	GPS REDACTED		608374795.716654	...	...	...	NULL																				
2	608320569.660951	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...			608374795.714319	...	...	...	NULL																				
3	608394313.376081	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...			608499172.123479	...	...	...	NULL																				
4	608394190.752453	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...			608499172.134406	...	...	...	608395693.418																				
5	608395679.09401	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...	-180.0	-180.0	608499172.119809	...	...	...	608395745.921																				
6	608394454.346502	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...	-180.0	-180.0	608499172.107273	...	...	...	608395653.764																				
7	608394265.038623	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...	-180.0	-180.0	608499172.114163	...	...	...	608395571.298																				
8	608394244.130864	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...	-180.0	-180.0	608499172.121111	...	...	...	608395533.538																				
9	608745299.077788	...	...	...	...	...	...	...	0.0	...	...	...	608745343	...	-180.0	-180.0	608745343.518228	...	...	...	NULL																				
10	608744441.23607	...	...	...	...	...	...	...	0.0	...	...	...	608744514	...	-180.0	-180.0	608744514.72861	...	...	...	NULL																				
11	608397114.048175	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...	-180.0	-180.0	608499172.117346	...	...	...	NULL																				
12	608395687.8371	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.121731	...	...	...	NULL																				
13	608394454.507178	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.136168	...	...	...	NULL																				
14	608394265.174128	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.13242	...	...	...	NULL																				
15	608394244.263782	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.100999	...	...	...	NULL																				
16	608390962.06688	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.128353	...	...	...	NULL																				
17	608390934.611421	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.048821	...	...	...	NULL																				
18	608390849.826394	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.041915	...	...	...	NULL																				
19	608390827.949491	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.045517	...	...	...	NULL																				
20	608390791.064269	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.039756	...	...	...	NULL																				
21	608390773.82038	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.050375	...	...	...	NULL																				
22	608390726.660095	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.033389	...	...	...	NULL																				
23	608390700.369266	...	...	...	...	...	...	...	0.0	...	...	...	608479437	...	-180.0	-180.0	608499172.130567	...	...	...	NULL																				
24	608389105.156867	...	...	...	...	...	...	...	0.0	...	...	...	NULL	...	-180.0	-180.0	608410273.5982	...	...	...	NULL																				
25	608349432.187898	...	...	...	...	...	...	...	7.73834467120181	...	...	...	NULL	...	-180.0	-180.0	608374795.718761	...	...	...	NULL																				

Figure 23. Photos.sqlite, table ZGENERICASSET: ZADDEDDATE, ZDURATION, ZLASTSHARED, latitude and longitude, ZMODIFICATIONDATE and ZTRASHEDDATE. Coordinates are redacted for the first four rows; -180.0 is a sentinel for no location.

Database Structure Browse Data Edit Pragmas Execute SQL						
SQL 1 x						
1	select datetime (creation_date+978307200, 'unixepoch','localtime') as timestamp,					
2	name, manufacturer, model, hardware, software from source_devices					
	timestamp	name	manufacturer	model	hardware	software
1	2020-03-21 18:33:20	iPhone	Apple Inc.	iPhone	iPhone8,4	13.3.1
2	2020-03-21 18:33:20	__NONE__				
3	2020-03-21 16:28:48	Apple Watch	Apple Inc.	Watch	Watch4,3	6.1.3
4	2020-03-21 16:28:48	__NONE__				
5	2020-03-27 15:06:14	AirPods	Apple Inc.	0x2002		
6	2020-03-31 15:08:08	AirPods Pro	Apple Inc.	0x200e		
7	2020-04-13 15:26:44	iPhone	Apple Inc.	iPhone	iPhone8,4	13.4.1
8	2020-04-13 21:16:00	Apple Watch	Apple Inc.	Watch	Watch4,3	6.2.1

Figure 24. Health database query on source_devices converting creation_date with the localtime modifier (examiner machine time zone, see Section 3.4).

	name	bundle_id	source_options	local_device	product_type
1	This Is's iPhone	com.apple.health.284D8298-B83A-422F-B075-A106BF2469F2	2	1	iPhone8,4
2	Health	com.apple.Health	3	0	
3	This Is's Apple Watch	com.apple.health.8DDD238A-F8E1-41D2-B0BC-D2756E655890	2	0	Watch4,3
4	This Is's Apple Watch	com.apple.health.9676B92F-913B-4B9D-AF56-5029C3E5FB64	2	0	Watch4,3
5	This Is's Apple Watch	com.apple.health.9676B92F-913B-4B9D-AF56-5029C3E5FB64	2	0	Watch4,3
6	Sync Solver	com.redshiftdev.Fitbit-Health-Sync	5	0	

Figure 25. Health database, table sources: six data sources including the iPhone, Apple Watch entries and the Fitbit Health Sync app.

	Energy Burned Goal Date	cache_index	energy_burned	energy_burned_goal	active_hours	active_hours_goal	brisk_minutes	brisk_minutes_goal	steps	walk_distance
1	0000-12-30 05:00:00	2020-03-20 04:00:00	0.0	0.0	0.0	12.0	0.0	30.0	0.0	0.0
2	2020-03-22 04:51:53	2020-03-21 04:00:00	28.965	480.0	1.0	12.0	0.0	30.0	54.0	45.6595000600815
3	2020-03-22 04:51:53	2020-03-22 04:00:00	454.2110000000001	480.0	15.0	12.0	28.0	30.0	5288.0	4141.56351345301
4	2020-03-22 04:51:53	2020-03-23 04:00:00	597.5709999999999	480.0	14.0	12.0	53.0	30.0	7930.0	6242.1125183487
5	2020-03-22 04:51:53	2020-03-24 04:00:00	525.601	480.0	14.0	12.0	40.0	30.0	5951.0	4704.22901471139
6	2020-03-22 04:51:53	2020-03-25 04:00:00	472.9620000000001	480.0	13.0	12.0	28.0	30.0	5560.0	4387.4990121531
7	2020-03-22 04:51:53	2020-03-26 04:00:00	564.0249999999999	480.0	15.0	12.0	41.0	30.0	6814.0	5433.98351609469
8	2020-03-22 04:51:53	2020-03-27 04:00:00	757.9350000000001	480.0	14.0	12.0	63.0	30.0	10094.0	8785.41520852542
9	2020-03-22 04:51:53	2020-03-28 04:00:00	673.801	480.0	14.0	12.0	44.0	30.0	10367.0	9052.75626033475
10	2020-03-22 04:51:53	2020-03-29 04:00:00	534.149	480.0	15.0	12.0	37.0	30.0	7762.0	6134.26842597011
11	2020-03-22 04:51:53	2020-03-30 04:00:00	881.534	480.0	16.0	12.0	77.0	30.0	12580.0	11142.2077181499
12	2020-03-22 04:51:53	2020-03-31 04:00:00	841.444	480.0	16.0	12.0	67.0	30.0	11119.0	9976.22446866435
13	2020-03-22 04:51:53	2020-04-01 04:00:00	576.732	480.0	15.0	12.0	44.0	30.0	7749.0	6229.00692761295
14	2020-03-22 04:51:53	2020-04-03 04:00:00	753.338	480.0	11.0	12.0	62.0	30.0	8787.0	8299.91420822666
15	2020-03-22 04:51:53	2020-04-02 04:00:00	812.379	480.0	14.0	12.0	65.0	30.0	10953.0	10099.5116853343
16	2020-03-22 04:51:53	2020-04-11 04:00:00	0.0	480.0	0.0	12.0	0.0	30.0	0.0	0.0
17	2020-03-22 04:51:53	2020-04-12 04:00:00	196.67	480.0	6.0	12.0	13.0	30.0	1633.0	1323.20716561802
18	2020-03-22 04:51:53	2020-04-13 04:00:00	599.029	480.0	14.0	12.0	35.0	30.0	6642.0	5423.58335220115
19	2020-03-22 04:51:53	2020-04-14 04:00:00	631.4399999999999	480.0	16.0	12.0	34.0	30.0	5970.0	4766.43763362939
20	2020-03-22 04:51:53	2020-04-15 04:00:00	625.1560000000001	480.0	14.0	12.0	37.0	30.0	7442.0	6007.9128400445
21	2020-03-22 04:51:53	2020-04-16 04:00:00	253.141	480.0	7.0	12.0	8.0	30.0	2657.0	2090.75641082722

Figure 26. healthdb_secure.sqlite, table activity_caches: daily energy, steps and walking distance values with UTC-converted dates.

	data_id	Start Date	End Date	data_type	activity_type	total_energy_burned	total_basal_energy_burned	total_distance	goal_type	goal
1	15017	2020-03-27 14:57:39	2020-03-27 15:15:39	79	37	208.57206077784	26.3355178817357	3.24798000599684	0	NULL
2	18146	2020-03-28 10:41:25	2020-03-28 10:58:44	79	37	203.107029922126	25.3377903447887	3.23536024568131	0	NULL
3	24649	2020-03-30 08:44:04	2020-03-30 09:10:35	79	37	302.580231726703	38.851842234397	4.84550446943786	0	NULL
4	30219	2020-03-31 14:58:47	2020-03-31 15:24:59	79	37	302.861406554144	38.3217232399215	4.86484843934411	0	NULL
5	36805	2020-04-02 13:36:41	2020-04-02 14:02:31	79	37	301.889197537766	37.9672273635361	4.85304426234419	0	NULL
6	44801	2020-04-03 15:11:28	2020-04-03 15:37:49	79	37	369.850771357538	46.0088782026736	4.86274976773959	0	NULL

Figure 27. Health database: workouts joined to samples with the localtime modifier, showing six running-type workouts (activity_type 37).

```
(kali@kali)~[~/var/mobile/Library/Health]
$ sqlite3 -header -csv healthdb_secure.sqlite "select datetime(cache_index+978307200,'unixepoch','utc') as Date, energy_burned as Calories from activity_caches;" > ~/Documents/Calories.csv
(kali@kali)~[~/var/mobile/Library/Health]
$
```

Figure 28. *sqlite3* command-line export of `cache_index` and `energy_burned` from `healthdb_secure.sqlite` to a CSV file in the examiner Documents folder.

Database Structure		Browse Data	Edit Pragmas	Execute SQL	
Create Table		Create Index	Modify Table	Delete Table	Print
Name	Type	Schema			
Tables (34)					
CdmaCellLocation	CREATE TABLE CdmaCellLocation				
CdmaCellLocationCounts	CREATE TABLE CdmaCellLocationCounts				
CdmaCellLocationLocal	CREATE TABLE CdmaCellLocationLocal				
CdmaCellLocationLocalCounts	CREATE TABLE CdmaCellLocationLocalCounts				
CellLocation	CREATE TABLE CellLocation (MCC				
CellLocationCounts	CREATE TABLE CellLocationCounts				
CellLocationLocal	CREATE TABLE CellLocationLocal				
CellLocationLocalCounts	CREATE TABLE CellLocationLocalCounts				
DatabaseIdentifier	CREATE TABLE DatabaseIdentifier				
DatabaseIdentifierCounts	CREATE TABLE DatabaseIdentifierCounts				
LteCellLocation	CREATE TABLE LteCellLocation (M				
LteCellLocationCounts	CREATE TABLE LteCellLocationCounts				
LteCellLocationLocal	CREATE TABLE LteCellLocationLocal				
LteCellLocationLocalCounts	CREATE TABLE LteCellLocationLocalCounts				
MicroLocationConfiguration	CREATE TABLE MicroLocationConf				
MicroLocationConfigurationCounts	CREATE TABLE MicroLocationConf				
MicroLocationMeasurements	CREATE TABLE MicroLocationMea				
MicroLocationMeasurementsCounts	CREATE TABLE MicroLocationMea				
MicroLocationModels	CREATE TABLE MicroLocationMod				
MicroLocationModelsCounts	CREATE TABLE MicroLocationMod				
MicroLocationRecordingEvents	CREATE TABLE MicroLocationRec				
MicroLocationRecordingEventsCounts	CREATE TABLE MicroLocationRec				
ScdmaCellLocation	CREATE TABLE ScdmaCellLocation				
ScdmaCellLocationCounts	CREATE TABLE ScdmaCellLocation				
TableInfo	CREATE TABLE TableInfo (TableNa				
VO2MaxHistory	CREATE TABLE VO2MaxHistory (id				
VO2MaxInputHistory	CREATE TABLE VO2MaxInputHist				
VO2MaxSummaryHistory	CREATE TABLE VO2MaxSummary				
WifiAWD	CREATE TABLE WifiAWD (Day INT				
WifiLocation	CREATE TABLE WifiLocation (MAC				
WifiLocationCounts	CREATE TABLE WifiLocationCount				
WifiTileBlobs	CREATE TABLE WifiTileBlobs (Tile				
WifiTileHeader	CREATE TABLE WifiTileHeader (Tile				
sqlite_sequence	CREATE TABLE sqlite_sequence(na				

Figure 29. *locationd cache_encryptedB.db* schema: 34 tables including `CellLocation`, `LteCellLocation` and `MicroLocation` tables (structure only; no rows reviewed).

Appendix C Queries and Commands

Commands and queries visible in the exhibits are reproduced below. Paths are relative to the extracted tree.

Purpose	Command or query
Retrieve the package	wget https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip
Hash the package	hashdeep ios_13_4_1.zip
Extract the file system	tar -xvf 13-4-1.tar
Render a plist	plistutil PublicEffectiveUserSettings.plist; plistutil IconState.plist
Health source devices	select datetime(creation_date+978307200,'unixepoch','localtime') as timestamp, name, manufacturer, model, hardware, software from source_devices
Health activity caches	select datetime(energy_burned_goal_date+978307200,'unixepoch','utc'), datetime(cache_index+978307200,'unixepoch','utc'), energy_burned, active_hours, brisk_minutes, steps, walk_distance from activity_caches
Health workouts	select ... datetime(samples.start_date+978307200,'unixepoch','localtime'), datetime(samples.end_date+978307200,'unixepoch','localtime') ... from workouts join samples on workouts.data_id = samples.data_id
Health CSV export	sqlite3 -header -csv healthdb_secure.sqlite "select datetime(cache_index+978307200,'unixepoch','utc') as Date, energy_burned as Calories from activity_caches;" > ~/Documents/Calories.csv
Cocoa to Unix conversion (manual)	unix = cocoa + 978307200; sms.db message.date: unix = (ns / 1e9) + 978307200

Where an exhibit truncates a query, the ellipsis above marks the omission; the full text should be copied from the saved SQL tab. Column names in the workouts query are not fully legible in the exhibit.

Appendix D Source Paths

Paths are given as cited in the brief; locations may appear under /var or /private/var because of symbolic links.

Artefact family	Path examined	Examined in exhibits?
Package settings	/var/mobile/Library/UserConfigurationProfiles/PublicInfo/PublicEffectiveUserSettings.plist	Yes (Figure 4)
Home screen	/var/mobile/Library/SpringBoard/IconState.plist	Yes
App containers	/var/mobile/Containers/Data/Application/<GUID> (listed); /var/containers/Bundle/Application	Listed only
Messages	/private/var/mobile/Library/SMS/sms.db	Yes
Contacts	/var/mobile/Library/AddressBook/AddressBook.sqlitedb	Yes (ABPerson)
Call history	/var/mobile/Library/CallHistoryDB/CallHistory.storedata	Yes

Artefact family	Path examined	Examined in exhibits?
Voicemail	/var/mobile/Library/Voicemail/voicemail.db	Yes (empty)
Calendar	/var/mobile/Library/Calendar/Calendar.sqlitedb; Extras.db	Structure only
Safari	/var/mobile/Library/Safari/History.db, Bookmarks.db, BrowserState.db, CloudTabs.db	Yes
Photos	/var/mobile/Media/DCIM/100APPLE; /var/mobile/Media/PhotoData/Photos.sqlite	Yes
Health	/var/mobile/Library/Health/healthdb.sqlite; healthdb_secure.sqlite	Yes (database per query to confirm)
Location	/var/root/Library/Caches/locationd/cache_encryptedB.db; /var/mobile/Library/Caches/com.apple.routined/Local.sqlite	Structure only
Device and SIM	/var/installd/Library/MobileInstallation/LastBuildInfo.plist; /private/var/wireless/Library/Preferences/com.apple.commcenter.plist; /private/var/logs/AppleSupport/general.log	Not in exhibits
Accounts and settings	/var/mobile/Library/Accounts/Accounts3.sqlite; com.apple.mobile.lidbackup.plist; com.apple.locationd.plist	Not in exhibits
KnowledgeC	/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db	Not in exhibits
Jailbreak indicators	checkra1n.dmg, Cydia-related artefacts and similar	Not in exhibits

Appendix E Tool Validation

Tools recorded in the exhibits: hashdeep, tar, DB Browser for SQLite, plistutil, sqlite3 and the Kali file manager. No Cellebrite Physical Analyzer or Magnet Axion output is included, so no tool-generated result could be compared with a manual one.

Validation performed. The Health activity_caches table was read through DB Browser for SQLite (Figure 26) and queried again through the sqlite3 command line (Figure 28). The CSV output was not displayed, so a match is **not yet demonstrated**. **[Open Calories.csv and compare at least one row with Figure 26 (for example 2020-03-22, 454.211 kcal), then record: matched, differed or could not be verified]**

[If Cellebrite Physical Analyzer or Magnet Axion was used, record the tool name, version, evidence source loaded and artefact category, and compare one timeline entry with its SQLite or plist record]

Appendix F Open Items Before Submission

The brief requires every core task to be covered. The exhibits compiled for this draft do not show the items below. Each is marked in yellow in the relevant section.

Task	Item to complete	Where
A	Serial, build, ICCID, IMSI, MCC, MNC; iCloud, backup, Find My and time-setting evidence; jailbreak search result	Sections 4.1, 4.2, 4.5
B	Calendar row counts; ABMultiValue lookup to match one contact or number to a call or message	Sections 5.2, 5.3
C	Bundle-versus-sandbox mapping for at least one app; photo and video counts from ZGENERICSASSET	Sections 4.4, 5.5
D	KnowledgeC streams; location rows or documented absence; Health queries re-run with the utc modifier	Sections 5.6, 5.7, 5.8, 3.4
E	Tool-versus-manual comparison; tool versions	Appendix E, Section 3.1
All	Name, registration number, cohort, date, declaration and assistance statement; final word-count check (2,500 to 3,500 words excluding cover, tables, figures and appendices)	Cover; declaration